ICASSP2026 Barcelona · Georgia Institute of Technology · NYU COURANT INSTITUTE SCHOOL OF MATHEMATICS, COMPUTING, AND DATA SCIENCE · MOHAMED BIN ZAYED UNIVERSITY OF ARTIFICIAL INTELLIGENCE · ByteDance

Demo Page · Paper

WHEN NOISE LOWERS THE LOSS: RETHINKING LIKELIHOOD-BASED EVALUATION IN MUSIC LARGE LANGUAGE MODELS

Xiaosha Li[1] · Chun Liu[2] · Ziyu Wang[3 4]

xiaosha@gatech.edu, chun.liu@bytedance.com, ziyu.wang@nyu.edu

[1]Georgia Institute of Technology · [2]ByteDance Inc. · [3]Courant Institute of Mathematical Sciences, NYU · [4]MBZUAI

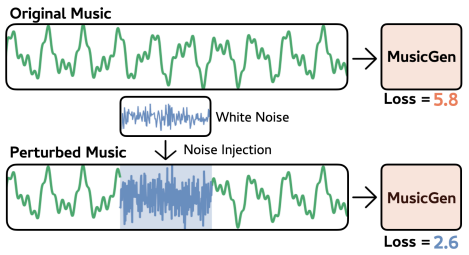
Paradox

Adding white noise to music reduces cross-entropy loss — the very metric used to evaluate music LLMs. This "Context Amnesia Effect" undermines loss as a quality indicator.

Motivation

- Music LLMs are commonly evaluated using cross-entropy loss — lower loss implies higher quality.
- Likelihood-based evaluation is known to suffer from exposure bias in NLP, but its failure modes in music remain unexplored.
- Key question: Can loss reliably distinguish high-quality music from "garbage music"?

Experiment Setup



- Models: MusicGen Small / Medium / Large / Melody, YuE (1B)
- Data: Training set · Generated · OOD classical (ASAP)
- White noise injected at token 250 (5 s), lengths: 0.1 – 4.0 s

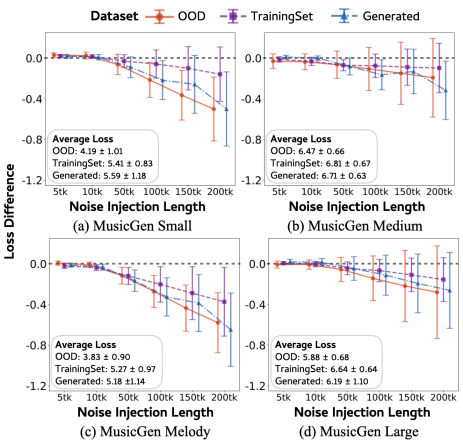
Loss definition

$$\ell(x_{1:T}) = -\sum_{t=1}^{T} \log p_\theta(x_t \mid x_{<t})$$

Perturbed signal

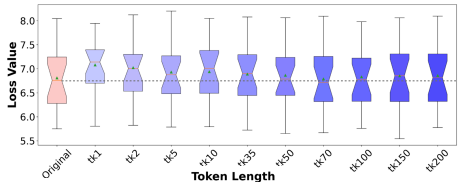
$$x'_i = \begin{cases} x_i, & i \notin \mathcal{I} \\ \epsilon_i, & i \in \mathcal{I} \end{cases}$$

Loss Difference vs. Noise Length



- As noise length increases, loss difference becomes more negative across all models and datasets.

Order Shuffling

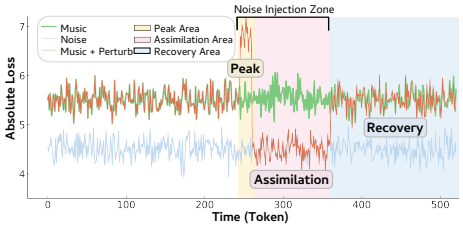


- Global loss stays flat even under 200-token shuffles — the model cannot detect long-range structural degradation.

Token-wise loss difference

$$\Delta\ell_t = -\log p_\theta(x'_t|x'_{<t}) + \log p_\theta(x_t|x_{<t})$$

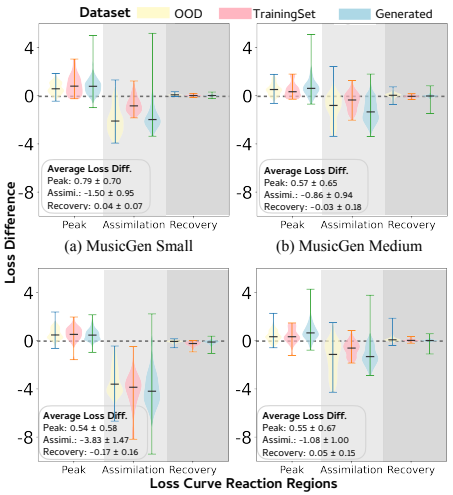
Context Amnesia Effect



Three labeled regions with brief descriptions:

- ① **Peak** (~100 ms) — loss spikes; model detects local inconsistency
- ② **Assimilation** — loss drops; model forgets context, adapts to noise
- ③ **Recovery** — loss returns but with high variance; exposure bias

Quantitative Results



- Small table with Peak / Assimilation / Recovery values for each model
- Moving average applied to $\Delta\ell_t$ to suppress local fluctuations; boundaries detected where loss crosses zero for **5 consecutive tokens** → robustly identifies Peak onset, Assimilation plateau, and Recovery phase.

Why Does This Happen?

- Noise produces out-of-distribution tokens → brief spike → rapid adaptation.
- Model shortens usable context, relies on corrupted input.
- Musical surprise and novelty may be penalized the same way as noise.

Training (token-level CE) ≠ Inference (sequential context) → Exposure Bias (cannot recover from errors → loss unreliable)

Proposed Direction — Profile-Based Evaluation

- The **shape** of the loss curve encodes quality, not its absolute value
- Strong Peak response = model has learned local musical integrity
- Label-free · Model-intrinsic · No annotations needed

Takeaways

- Cross-entropy loss alone is **unreliable** as a music quality metric
- The **Context Amnesia Effect** is consistent across all 5 models
- Music LLMs detect **local texture** but are blind to **global structure**
- Loss curve **shape** — especially the onset Peak — is the way forward

Conclusion

- Current LLMs cannot reliably use absolute loss to distinguish between canonical classical compositions and generic training-set samples.
- In music, surprise, tension, and novelty are essential — unfamiliar yet meaningful passages may be misinterpreted as perturbations and undervalued the same way as noise.

The shape of the loss curve encodes musical quality — not its absolute value.

Acknowledgment

- The authors thank Prof. Alexander Lerch, Yikang Shen, and Wenye Ma for their valuable guidance.